The Pylyavskyy–Yang hexagon is the vertex star

Q569: the hexagonal puzzle piece is a *bulk* tile, and its boundary piece is a *truncation*, not a K-matrix

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Provenance (PROTOCOL §2.3).
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Repository: `clio-vega/proofs`. **Commit:** `18e277a5d0` in `clio-vega/proofs` — the commit of the sources this PDF was built from (this `.tex` and the scripts in `proofs/code-1023c2-Q569/`, committed together). The PDF itself is added in the immediately following commit.
Source read: `library/2026-10-23-browse-arxiv-raw/ep-1801.07667/` (`ktiles.tex`, 1725 lines; `hex-tiling.tex`, 65 lines), e-print of Pylyavskyy–Yang, arXiv:1801.07667. Extraction level: `deep-read` (tile macros, all four tiling theorems, §`ss:K-coefficients`, §`ss:protrude` read verbatim in the source).
Status: the geometric theorems are `proved`; the §`ss:protrude` equivalence is `computed` (census domain stated in §8); the integrability *negatives* are `proved` where stated as such and flagged otherwise.

Abstract

Pylyavskyy and Yang (arXiv:1801.07667) introduce two hexagonal puzzle pieces $\mathsf{hex}, \mathsf{hex}^{\mathsf{R}}$ giving Littlewood–Richardson rules for the K-homology of Grassmannians. Their paper contains no integrability claim; the question Q569 asked whether hex is a *bulk* R-matrix piece or a *boundary* K-matrix piece satisfying a reflection equation, the latter suggested by §`ss:protrude`, where a distinguished trapezoidal tile is introduced and explicitly forbidden outside the left edge of the puzzle.

The answer is *neither*, and the reason is a single geometric identification: **hex and $\mathsf{hex}^{\mathsf{R}}$ are the two orientation-keyed alternating labellings of the closed vertex star of the triangular lattice** (Theorem 4) — the union of the six unit triangles containing a lattice vertex. Everything the paper observes follows. The hexagon is a bulk piece indexed by its centre; it fails to fit inside a wall exactly when its centre *lies on* the wall, and then exactly half of it is outside (Theorem 7), which is why enlarging the ambient rectangle by one suffices; the §`ss:protrude` trapezoid is *literally* the hexagon intersected with the half-plane, with every retained edge label unchanged (Theorem 9); and $\mathsf{hex}^{\mathsf{R}}$ is the image of hex under every element of D_6 that exchanges the two triangle orientations (Corollary 5), so the two theorems are one theorem conjugated by Grassmannian duality, not two boundaries of one model.

The reflection-equation reading is refuted on three independent grounds (§6): the boundary piece has *zero moduli*, being determined by the bulk piece and the wall; the bulk tile set is *chiral*, so no lattice reflection preserves it and there is nothing to reflect in; and the third edge label 2 is a deterministic wall-adjacency indicator rather than an amplitude. Separately, the tiles are not vertices of a line-conserving model at all (§7): ten of the seventeen carry exactly one 1-edge in each of the three direction classes.

All of this rests on a tile extraction verified by four independent positive controls (§3): an engine built from set-valued tableaux reproduces the puzzle counts for all four of the paper's

theorems (208/208, 208/208, 187/187, 54/54 at $n = 5, k = 2$; 875/875 at $n = 6, k = 3$). I also verify the paper's *unproved* §`ss:protrude` claim (187/187 and 875/875), and quantify the protrusion loss on the paper's own example: $d = 4$, the unfixed count is 2, and both of the paper's fixes give 4.

1 The question, and the answer

Q569, as posed. Is the hex/hex^{R} tile set a *bulk* R-matrix satisfying the Yang–Baxter equation, or a bulk R-matrix *plus* a one-sided boundary K-matrix satisfying the reflection (Sklyanin) equation?

The evidence for the second reading was three sentences of §`ss:protrude`: "the hex piece will protrude to the left of the puzzle with side length 7"; "(By the way things are set up, the hexagon never needs to protrude to the right or below.)"; and, of the alternative fix by an extra trapezoidal piece, "However, we do not want this piece used elsewhere." A tile that acts at one edge and is forbidden in the interior does look like a K-matrix.

Answer. Neither. The hexagon is a bulk piece. The trapezoid is a boundary piece, but it is the bulk piece *clipped by the wall*, and a clipping is not a reflection: it has no spectral parameter, no amplitude and no moduli. The one-sidedness that motivated the question is real, and Theorem 7 explains it exactly — but it is a statement about where a vertex star can sit relative to a lattice line, not about an integrable boundary condition.

2 Objects: the tiles, extracted

Everything below is extracted from the e-print's own `tikz` macros (`ktiles.tex`, lines 232–236):

$$\texttt{\textbackslash r\{i\}} = \text{unit edge, direction } 60i^\circ, \text{ label 0}; \quad \texttt{\textbackslash b\{i\}} = \text{label 1}; \quad \texttt{\textbackslash bb\{i\}} = \text{label 2}.$$

Since `\xcoord` $= \cos 60^\circ$ and `\ycoord` $= \sin 60^\circ$ (`ktiles.tex`, lines 4–5), the paper's `tikz` coordinates *are* the triangular-lattice basis. I therefore work throughout in integer coordinates $(a, b) \leftrightarrow ae_0 + be_1$ with $e_0 = (1, 0)$, $e_1 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$, so that

$$D_0, \ldots, D_5 \;=\; (1,0),\ (0,1),\ (-1,1),\ (-1,0),\ (0,-1),\ (1,-1).$$

The shear to Euclidean coordinates is affine, so incidence, convexity and point-in-polygon are unaffected.

A tile is a closed counterclockwise walk of unit steps carrying labels. The seventeen tiles of the paper are then, verbatim:

tile	source	step sequence (dir, label)	#△
0-triangles	fig. LR-tiles	$(0,0)(2,0)(4,0)$; $(1,0)(3,0)(5,0)$	1
1-triangles	fig. LR-tiles	$(0,1)(2,1)(4,1)$; $(1,1)(3,1)(5,1)$	1
rhombi	fig. LR-tiles	$(0,1)(1,0)(3,1)(4,0)$, + 2 rotations	2
K (Vakil)	fig. K-tiles	$(1,1)(1,0)(3,1)(3,0)(5,1)(5,0)$	4
tri (WZ)	fig. K-tiles	$(0,1)(0,0)(2,1)(2,0)(4,1)(4,0)$	4
hex	fig. K-tiles	$(1,0)(2,1)(3,0)(4,1)(5,0)(0,1)$	6
hex^{R}	fig. K-tiles	$(0,0)(1,1)(2,0)(3,1)(4,0)(5,1)$	6
trap	§`ss:protrude`	$(1,0)(2,1)(4,0)(4,1)(0,1)$	3

together with the five relabelled pieces of §`ss:protrude` (§8). All seventeen close and have positive signed area (verified).

Remark 1 (the alternation)**.** The KTW rhombi carry labels $(1,0,1,0)$ around their boundary and $\mathsf{hex}, \mathsf{hex}^{\mathsf{R}}$ carry $(0,1,0,1,0,1)$: both are *alternating*. The trapezoid carries $(0,1,0,1,1)$ — alternation broken. So the hexagon is the $m=3$ member of the family whose $m=2$ member is the KTW rhombus, and the trapezoid is not a member of that family at all. This is the first sign that the trapezoid is a derived object rather than a new primitive.

Convention 2 (boundary, verified against the paper's own figure)**.** Let $n \geq k$ and let the ambient rectangle be k rows of length $n-k$. For a partition λ inside it, $\sigma(\lambda) \in \{0,1\}^n$ is the lower-right boundary path read from top right to bottom left, 0 for horizontal and 1 for vertical (`ktiles.tex`, lines 526–536). The puzzle region is the upright triangle with vertices $(0,0),(n,0),(0,n)$, and

$$\begin{aligned} \text{bottom, } (0,0)\to(n,0) \text{ in direction } 0: &\ \sigma(\nu), \\ \text{left, } (0,0)\to(0,n) \text{ in direction } 1: &\ \sigma(\lambda), \\ \text{right, } (0,n)\to(n,0) \text{ in direction } 5: &\ \sigma(\mu). \end{aligned}$$

I verified all three against figure `bdry` (`ktiles.tex`, lines 610–617), which prints the three sides explicitly for $\lambda=(2,1,0), \mu=(3,2,0), \nu=(4,3,1)$, $n=9$, $k=3$: exact match on all 27 edges.

Remark 3 (the wall)**.** The *left* side of the puzzle region runs from $(0,0)$ to $(0,n)$, i.e. in direction 1 (equivalently 4). This is the side past which the hexagon protrudes, and it is exactly the direction class on which §`ss:protrude` introduces the third label 2. I call the line $a=0$ the *wall* W.

3 Positive controls

The brief's first instruction was to confirm a well-defined object exists before testing any equation on it. I built two engines and made them disagree or agree.

Engine A (puzzles). Exact cover of the n^2 unit triangles of the region by lattice translates of the tiles, with unit-edge labels matching between adjacent tiles and against Convention 2. Edges interior to a tile are unconstrained.

Engine B (the ring, independent). The single stable Grothendieck polynomial $G_\lambda = \sum_T (-1)^{|T|-|\lambda|} x^T$ over semistandard *set-valued* tableaux (`ktiles.tex`, lines 711–716), multiplied and comultiplied as an actual polynomial, then expanded in the G-basis by degree-triangular peeling ($G_\lambda = s_\lambda + \text{higher}$). Engine B touches no puzzle and no tableau rule from the paper: it goes through the ring. Controls on Engine B alone: the minimal-degree part of G_λ is s_λ; and $c^\nu_{\lambda\mu}$ at $|\nu| = |\lambda|+|\mu|$ reproduces the classical Littlewood–Richardson coefficients of s^2_{21}.

check	domain	result
KTW count $= c^\nu_{\lambda\mu}$, paper's running example	n=9, k=3, $(2,1),(3,2),(4,3,1)$	$2=2$
K-puzzles $= c^\nu_{\lambda\mu}$ (K-theory, Vakil)	n=5, k=2, all (λ,μ,ν)	208/208
tri-puzzles $= \tilde{c}^\nu_{\lambda\mu}$ (Wheeler–Zinn-Justin)	n=5, k=2, all	208/208
hex-puzzles $= d^\nu_{\lambda\mu}$ (K-homology)	n=5, k=2, all	187/187
$\mathsf{hex}^{\mathsf{R}}$-puzzles $= d^\nu_{\lambda\mu}$	n=5, k=2, all	54/54
hex-puzzles $= d^\nu_{\lambda\mu}$, one level up	n=6, k=3, all	875/875

Census domain, stated exactly. "All (λ,μ,ν)" means: every triple of partitions inside the ambient rectangle prescribed by the relevant theorem — $k\times(n-k)$ for K, tri; $k\times(n-k-1)$ for hex; $(k-1)\times(n-k)$ for $\mathsf{hex}^{\mathsf{R}}$ — subject to the theorem's degree inequality ($|\nu| \geq |\lambda|+|\mu|$ for K, tri; $|\nu| \leq |\lambda|+|\mu|$ for $\mathsf{hex}, \mathsf{hex}^{\mathsf{R}}$). The $n=6, k=3$ row is the "one level up" test: it moves k off 2, where a two-row coincidence could not be detected.

4 The hexagon is the vertex star

Theorem 4 (Vertex star)**.** *Let v be a vertex of the triangular lattice and let* $\mathrm{St}(v) = \bigcup\{T : T \text{ a unit triangle}, v \in T\}$ *be its closed star, the union of the six unit triangles containing v. Then:*

(i) $\mathrm{St}(v)$ *is the convex hexagon with vertex set* $\{v + D_j\}_{j\in\mathbb{Z}/6}$*, and its boundary is the closed hexagonal walk* $v + D_0 \to v + D_1 \to \cdots \to v + D_5 \to v + D_0$ *of six unit edges.*

(ii) The six triangles of $\mathrm{St}(v)$ *alternate in orientation around v, and the boundary edge contributed by each is the edge of that triangle opposite v.*

(iii) There are exactly two alternating $\{0, 1\}$*-labellings of this hexagonal boundary. They are* hex *and* $\mathsf{hex}^{\mathsf{R}}$*, and they are distinguished by triangle orientation:* hex *puts the label* 1 *on the outer edge of each* upward *triangle and* 0 *on each* downward *one;* $\mathsf{hex}^{\mathsf{R}}$ *reverses this.*

Proof. (i) The unit triangles containing v are exactly $T_j = \{v,\ v+D_j,\ v+D_{j+1}\}$ for $j \in \mathbb{Z}/6$, since consecutive D_j, D_{j+1} are at 60° and span a unit triangle with v, and every unit triangle containing v arises this way. Their union is the convex hull of $\{v + D_j\}$, and the edges $[v + D_j, v + D_{j+1}]$ not containing v concatenate to the stated closed walk.

For the identification with the paper's tiles: summing the step sequences, hex and $\mathsf{hex}^{\mathsf{R}}$ each close, and in each case the six vertices minus their centroid are precisely $D_0, \ldots, D_5$ (computed exactly: the vertex coordinate sums are divisible by 6, so the centroid is a lattice point v, and the vertex set is $v + D$). Hence both tiles equal $\mathrm{St}(v)$ as regions.

(ii) T_j and T_{j+1} share the edge $[v, v + D_{j+1}]$ and lie on opposite sides of it, so they have opposite orientations; the parity of j therefore determines the orientation class. The boundary edge of T_j is $[v + D_j, v + D_{j+1}]$, which is the edge of T_j not meeting v.

(iii) An alternating labelling of a 6-cycle is determined by its value on one edge, so there are exactly two. By (ii) the two alternation classes of edges are precisely the two orientation classes of triangles, so the two labellings are the two orientation-keyed ones. Reading the paper's step sequences (§2) and matching each boundary edge to its triangle gives: in hex the upward triangles' outer edges carry 1 and the downward ones 0; in $\mathsf{hex}^{\mathsf{R}}$ the reverse (verified directly). □

Corollary 5 ($\mathsf{hex}^{\mathsf{R}}$ is hex seen in the other orientation)**.** *Let $g \in D_6$ act on the lattice. Then* $g(\mathsf{hex}) = \mathsf{hex}^{\mathsf{R}}$ *if and only if g exchanges the two triangle orientations; otherwise* $g(\mathsf{hex}) = \mathsf{hex}$*. Explicitly, the odd rotations* ρ^1, ρ^3, ρ^5 *and the three reflections whose axes lie* along *edge directions exchange them; the even rotations and the three reflections whose axes lie at* 30° *to the edges do not.*

Proof. Immediate from Theorem 4(iii), since the labelling is a function of triangle orientation and g maps $\mathrm{St}(v)$ to $\mathrm{St}(gv)$. The classification of which g exchange orientations: writing a reflection as $d \mapsto m - d$ on direction indices, it fixes a direction iff $2d \equiv m \pmod 6$, which is solvable iff m is even; the m even reflections (axis along an edge direction) exchange up and down triangles and the m odd ones do not (verified on $\mathrm{up}(0, 0)$ for all twelve group elements). □

Remark 6. Corollary 5 is the structural content of the paper's two hexagonal theorems. It is *not* that hex and $\mathsf{hex}^{\mathsf{R}}$ are two boundaries of one model; they are one tile in the two orientation conventions. Consistently, the paper *defines* $\tilde{d}^{\nu}_{\lambda\mu} := d^{\nu'}_{\lambda'\mu'}$ on transposed partitions (`ktiles.tex`, lines 757–760), and the rectangle bookkeeping matches exactly: with $k^* = n - k$ and $n^* = n$, the $\mathsf{hex}^{\mathsf{R}}$ ambient rectangle $(k-1)\times(n-k)$ transposes to $(n-k)\times(k-1) = k^*\times(n^*-k^*-1)$, the hex

ambient rectangle. So Theorem `hexR-tiling` is Theorem `hex-tiling` conjugated by $k \mapsto n-k$, i.e. by the Grassmannian duality $\mathrm{Gr}(k,n) \cong \mathrm{Gr}(n-k,n)$.

5 Protrusion is a half-overlap, and the trapezoid is a clipping

Theorem 7 (Protrusion)**.** *Let W be a lattice line and let H^+, H^- be the two closed half-planes it bounds. For a lattice vertex v:*

$$\mathrm{St}(v) \subseteq H^+ \iff v \in H^+ \setminus W, \qquad \mathrm{St}(v) \subseteq H^- \iff v \in H^- \setminus W,$$

and $\mathrm{St}(v)$ meets both open half-planes if and only if $v \in W$. In that case exactly three of its six triangles lie in each half-plane, and $W \cap \mathrm{St}(v)$ is the segment $[v-e, v+e]$ of length 2, where e is the unit vector along W.

Proof. Choose the basis so that $W = \{a = 0\}$ and $e = e_1$ (possible since W is a lattice line, hence parallel to some D_j; take $e_1 = D_j$). The six neighbours $v + D_j$ of $v = (a_v, b_v)$ have a-coordinates $a_v + (1, 0, -1, -1, 0, 1)$, so

$$\mathrm{St}(v) \subseteq \{\, a_v - 1 \le a \le a_v + 1 \,\},$$

with both extremes attained. Hence $\mathrm{St}(v) \subseteq \{a \ge 0\}$ iff $a_v \ge 1$, and $\mathrm{St}(v) \subseteq \{a \le 0\}$ iff $a_v \le -1$; the remaining case is $a_v = 0$. If $a_v = 0$ the six triangles T_j split by the sign of the a-coordinates of their non-v vertices: three have an $a = -1$ vertex and three an $a = +1$ vertex, and no triangle has both (consecutive D_j, D_{j+1} never have a-coordinates $+1$ and -1). So the split is $3 + 3$. Finally $W \cap \mathrm{St}(v)$ contains v and the two neighbours $v \pm e_1$ (the only neighbours on W), and is convex, so it is $[v - e_1, v + e_1]$. □

Corollary 8 (why "slightly smaller" is enough)**.** *Because the obstruction occurs only for $v \in W$, a hexagon can protrude past the wall by exactly one lattice step and no more. Enlarging the ambient rectangle by one in the direction normal to the wall moves every admissible centre to $a_v \ge 1$ and removes the obstruction. This is the paper's Theorem `hex-tiling` hypothesis and the footnote "For technical reasons, we require partitions to be slightly smaller."*

Theorem 9 (The trapezoid is a truncation)**.** *With $W = \{a = 0\}$ the wall and the puzzle in $\{a \ge 0\}$, let $v \in W$ so that $\mathrm{St}(v)$ straddles. Then the §`ss:protrude` trapezoid is exactly*

$$\mathsf{trap} \;=\; \mathsf{hex} \cap \{a \ge 0\},$$

as labelled tiles: the three retained boundary edges keep their labels unchanged, the three edges lying in $\{a < 0\}$ are discarded, and the only new boundary edges are the two unit edges of $W \cap \mathrm{St}(v) = [v - e_1, v + e_1]$, carrying label 0 on $[v, v + e_1]$ and label 1 on $[v - e_1, v]$.

Proof. Verified exactly, by comparing covered unit triangles and labelled boundary edges. With hex anchored so that $v = (0, 1)$: $\mathrm{St}(v)$ covers the six triangles $\{U(0,0), U(0,1), U(-1,1), D(0,0), D(-1,0), D(-1,1)$ of which exactly $\{U(0,0), U(0,1), D(0,0)\}$ lie in $\{a \ge 0\}$; the trapezoid covers precisely these three. The retained boundary edges are $[(1,0),(1,1)]$ with label 0, $[(1,1),(0,2)]$ with label 1, and $[(0,0),(1,0)]$ with label 1 — identical in hex and in trap. The discarded edges are $[(-1,2),\cdot]$ label 0, $[(-1,1),\cdot]$ label 1, $[(0,0),\cdot]$ label 0. The new edges are $[(0,1),(0,2)]$ label 0 and $[(0,0),(0,1)]$ label 1, which are the two halves of $W \cap \mathrm{St}(v)$ above and below $v = (0,1)$. The area accounting is consistent: $\#\mathsf{trap} = 3 = \frac{1}{2}\#\mathsf{hex}$, as Theorem 7 requires. □

Remark 10 (one bit of new data, and no freedom in it)**.** Clipping does introduce something not present in hex: the two cut edges were *interior* edges of the hexagon (the radii $[v, v \pm e_1]$), hence unlabelled, and become boundary edges requiring labels. Theorem 9 records that these are 0 above v and 1 below. This is the trapezoid's entire content beyond the hexagon — one bit — and it is not a free parameter: the paper lists a single trapezoid, and §8 verifies computationally that this single choice reproduces $d^{\nu}_{\lambda\mu}$.

6 There is no reflection to reflect in

A K-matrix is a boundary vertex that *reflects* bulk lines and satisfies the reflection equation $K_2 R_{21} K_1 R_{12} = R_{21} K_1 R_{12} K_2$. Pre-registered prediction P2 said the trapezoid plays this role. It does not, for three independent reasons.

Proposition 11 (Zero moduli)**.** *The boundary piece is determined by the bulk piece and the wall:* $\mathsf{trap} = \mathsf{hex} \cap H^+$ *(Theorem 9), and the clipping is unique once the wall is fixed (Theorem 7 gives a unique straddling configuration, with a unique cut). A K-matrix, by contrast, is an* independent *solution of the reflection equation: it carries its own spectral parameter and generically belongs to a positive-dimensional family for a given R. The PY boundary piece has no spectral parameter and no family.*

Proposition 12 (The bulk tile set is chiral)**.** *No reflection of the triangular lattice preserves the KTW tile set.*

Proof. The KTW rhombus rule reads: edges are labelled "0 if clockwise of an acute angle and 1 if clockwise of an obtuse angle" (`ktiles.tex`, line 577). A reflection reverses orientation, hence exchanges "clockwise" for "counterclockwise", and therefore sends each rhombus to the rhombus on the same pair of direction classes carrying the *complementary* labelling. None of the three complementary rhombi is a KTW piece. Verified for all six reflections: each sends all three rhombi outside the set (while permuting the 0- and 1-triangles among themselves). The six rotations, by contrast, all preserve the set. □

Proposition 13 (The hexagon is not wall-reflection-invariant)**.** *Let σ_W be the reflection in the wall W. As a* region*, $\mathrm{St}(v)$ is σ_W-invariant for $v \in W$. As a* labelled tile *it is not: $\sigma_W(\mathsf{hex}) = \mathsf{hex}^{\mathsf{R}}$.*

Proof. W is a lattice line, so σ_W has its axis along an edge direction; by Corollary 5 such a reflection exchanges the triangle orientations, hence exchanges hex and $\mathsf{hex}^{\mathsf{R}}$. Regionally σ_W fixes v and permutes its neighbours, so it fixes $\mathrm{St}(v)$. □

Remark 14 (what Proposition 13 costs the reflection reading)**.** This is the sharpest of the three. Under the wall reflection the hexagon does not return to itself but to $\mathsf{hex}^{\mathsf{R}}$ — a *different* tile, which computes a *different* coefficient ($\tilde{d}$ rather than d) in a *different* ambient rectangle. So the two halves of a straddling hexagon are mirror images as shapes but carry mismatched labels under the mirror: the clipping is not a fold, and there is no reflection symmetry at the wall for a reflection equation to express.

Proposition 15 (The label 2 is an indicator, not an amplitude)**.** *In the §`ss:protrude` tile set, a 1-edge in the wall's direction class $\{1, 4\}$ carries label 1 if and only if it lies on W, and label 2 if and only if it is interior.*

Proof. Counterclockwise traversal puts a tile's interior on the left of each directed edge. Hence a tile presents a direction-1 edge exactly when it lies to that edge's upper left, and a direction-4 edge exactly when it lies to its lower right. The paper relabels every 1-edge on a tile's lower-right side to 2 (so direction-1 1-edges are always 2: pieces `\bb1\b3\b5`, `\bb1\r2\b4\r5`, `\bb1\r2\bb4\r5`), and duplicates every piece with a 1-edge on its upper-left side (direction 4) into a label-1 and a label-2 version. Now let e be an interior lattice edge in this class: the tile above left of e presents it in direction 1, hence labels it 2, so the tile below right must also label it 2, i.e. must be the label-2 version. If instead $e \subseteq W$ there is no tile above left, and the tile below right labels it 1 — consistent with Convention 2, under which the wall carries $\sigma(\lambda) \in \{0,1\}^n$. So label 1 on this class occurs exactly on W. In particular trap, whose direction-4 1-edge carries label 1 with no label-2 counterpart, can be placed only at the wall — the paper's "the new trapezoid piece cannot be used except at the left boundary". There is no sum over boundary states anywhere in this: the label is a deterministic function of wall-adjacency. □

7 There is no vertex model on these tiles

Pre-registered prediction P1 said the bulk set does not satisfy a plain Yang–Baxter equation *for any assignment of spectral parameters and weights*. That prediction is not well posed: the tiles are faces of a tiling model, all of weight 1, and the paper's theorems count cardinalities; there is no R to test until one fixes a map from tiles to a matrix on a tensor product, and the paper supplies none. (This is exactly the failure the brief warned about — check that the objects exist before grading the chances.)

Here is the well-posed replacement. In a vertex model the edge states are line occupations, and a line entering a vertex must leave it, so each vertex carries an *even* number of occupied edges in each direction class. Taking the 1-edges as the lines and the classes $H = \{0,3\}$, $N = \{1,4\}$, $W' = \{2,5\}$:

Proposition 16. *Ten of the seventeen tiles violate line conservation. The* 0*-triangles and all rhombi conserve (counts* $(0,0,0)$ *and one class* $= 2$*); the* 1*-triangles,* K*,* tri*,* hex*,* $\mathsf{hex}^{\mathsf{R}}$*,* hex_2 *and* trap *each carry exactly* one 1*-edge in each of the three classes, i.e.* $(1,1,1)$ *— odd in all three.*

Proof. Direct count from the step sequences of §2. □

So the 1-edges form a graph in which rhombi are straight segments and every other nontrivial tile is a Y-junction: a branching structure, not a conserving one. The tile set is therefore not itself a vertex model, and the question "does it satisfy YBE" has no object. This is consistent with the paper's measured silence (the brief's census: *Yang–Baxter*, *R-matrix*, *integrable*, *vertex model*, *transfer matrix*, *lattice model* — 0 occurrences each across 1790 lines, against six positive controls firing on the same instrument).

Remark 17 (scope — what this does *not* say)**.** It does not say that Grassmannian puzzles are non-integrable. They are known to be integrable, and the paper's own tri piece is due to Wheeler–Zinn-Justin, who obtained it *from* an integrable lattice model. Proposition 16 says only that the integrable structure does not live on *these faces with these labels as line occupations*; any integrable formulation must pass through a different presentation, and Pylyavskyy–Yang supply none for the two new pieces. Whether the hexagon has an R-matrix provenance in some other presentation is open and is *not* settled here.

8 The §ss:protrude alternative, verified

The paper offers its second fix without a theorem or a proof: "Here is the complete set of tiles", followed by a figure of twelve pieces and three sentences of explanation. The claim is that with the refined alphabet $\{0, 1, 2\}$ one may use the *natural* side length rather than enlarging it. I tested it.

The twelve pieces are: the two 0-triangles; the 1-triangles `\b0\b2\b4`, `\b0\b2\bb4`, `\bb1\b3\b5`; the rhombi `\b0\r1\b3\r4`, `\b2\r3\b5\r0`, `\bb1\r2\b4\r5`, `\bb1\r2\bb4\r5`; and the green pieces `\r1\b2\r3\b4\r5\b0` (= hex), `\r1\b2\r3\bb4\r5\b0`, `\r1\b2\r4\b4\b0` (= trap).

Proposition 18 (computed)**.** *For all λ, μ, ν inside the ambient rectangle $k \times m$, with $|\nu| \leq |\lambda| + |\mu|$, the number of refined-set tilings at side $n = k + m$ equals the number of* hex*-tilings at side $n + 1$ equals $d^{\nu}_{\lambda\mu}$.* Census domain: $(k, m) = (2, 2)$*, all* 187 *admissible triples,* 44 *of them with $d^{\nu}_{\lambda\mu} \neq 0$; and $(k, m) = (2, 3)$, all* 875 *admissible triples,* 143 *of them with $d^{\nu}_{\lambda\mu} \neq 0$. Agreement* 187/187 *and* 875/875 *for both the refined set and the enlarged* hex *set, against Engine B.*

I also ran the paper's own §ss:protrude example, $\lambda = (2, 1)$, $\mu = (4, 2)$, $\nu = (4, 3, 1)$, $k = 3$, which is where the protrusion is exhibited:

$d^{\nu}_{\lambda\mu}$ from Engine B (independent)	4
hex-puzzles at side 7 (the "natural" side; theorem's hypothesis fails, $\mu_1 = 4 > 3$)	2
hex-puzzles at side 8 (the theorem's prescribed side)	4
refined §ss:protrude set at side 7	4

So at the natural side length the plain hex set loses exactly two tilings — the protrusion, quantified — and both of the paper's fixes recover all four.

Corollary 19 (the shrink is not intrinsic)**.** *The one-unit shrink of the ambient rectangle is not a structural feature of the* hex *rule. It is one of two equivalent bookkeeping devices for the same clipping: move the wall away from the hexagon's centre, or clip the hexagon and mark wall-adjacency in the alphabet.*

9 The pre-registered predictions, graded

P1 *The bulk set does not satisfy a plain YBE for any assignment of spectral parameters and weights. Predicted: confirmed.* **SPLIT.** The conclusion stands but the reason in the brief was wrong: there is no weight object to quantify over, so the statement as written is ill-posed. The well-posed replacement (Proposition 16) is stronger in one direction — there is no vertex model on these tiles at all — and weaker in another: it says nothing about other presentations.

P2 *With the trapezoid as K and the rhombi as R, the set satisfies a reflection equation. Predicted: confirmed; the risky arm.* **REFUTED**, on three independent grounds: zero moduli (Proposition 11), chirality of the bulk (Proposition 12), and $\sigma_W(\mathsf{hex}) = \mathsf{hex}^{\mathsf{R}}$ (Proposition 13). This is the arm I most wanted to be wrong about, and it died.

P3 *The one-unit shrink is the boundary site:* hex *and* $\mathsf{hex}^{\mathsf{R}}$ *are the two one-sided boundaries of the same bulk model. Predicted: confirmed.* **REFUTED as stated.** hex and $\mathsf{hex}^{\mathsf{R}}$ are one tile in the two triangle orientations (Corollary 5), and the second theorem is the first conjugated by $k \mapsto n - k$ with transposed partitions. There is one boundary phenomenon seen twice through duality, not two boundaries. What survives: the protrusion really is one-sided, and Theorem 7 says why. What additionally dies: the shrink is not intrinsic (Corollary 19).

P4 *The degree-inequality flip is the K-homology/K-theory duality and not anything about the boundary — i.e. P4 predicts this clue is a red herring for Q569. Predicted: confirmed.* **CONFIRMED.** The flip is product versus coproduct: $|\nu| \geq |\lambda| + |\mu|$ for $G_\lambda G_\mu$ and $|\nu| \leq |\lambda| + |\mu|$ for $\Delta(G_\nu)$ (`ktiles.tex`, lines 725–728), both reproduced in Engine B. It is orthogonal to the wall.

Pre-committed falsifiers. "If the bulk tiles do satisfy a plain YBE": not reached, since no YBE object exists (P1 SPLIT). "If no consistent weight assignment exists at all": this is what happened, and I report it as Proposition 16 rather than as a refutation of the paper, which claims nothing about weights. "If P2 holds but only after the trapezoid is allowed in the bulk": not reached; P2 failed outright.

10 Gaps, precisely stated

1. **Proposition 18 is `computed`, not `proved`.** I verified the §`ss:protrude` equivalence on two ambient rectangles (2×2, 2×3) and on the paper's own example. I do not have a bijection. The natural candidate — "clip every hexagon whose centre lies on the wall, and read the extra leading 0 of $\sigma(\cdot, n+1, k)$ as a frozen strip of 0-triangles" — I checked only to the extent that the corner triangle $U(0,0)$ of the enlarged puzzle is forced to be a 0-triangle (the only tile admitting two 0-labelled boundary edges there). Propagating that to a full frozen region, and matching the clipped hexagons across the two models, is not done.

2. **The integrability question is settled only in the presentation I tested.** Proposition 16 rules out a line-conserving vertex model on these faces with these labels. It does not rule out an R-matrix for hex in another presentation, e.g. the one Wheeler–Zinn-Justin use for tri. I did not read [3]; I have not opened it, and no claim here depends on it.

3. **Proposition 11 contains a judgement, not a theorem.** "A K-matrix generically belongs to a positive-dimensional family" is standard but I state it from general knowledge, not from a source read this session. The hard part — that trap is determined by hex and W — is Theorem 9 and is proved.

4. **Census domains are small.** The four-theorem control reaches $n \leq 6$, $k \leq 3$; the §`ss:protrude` check reaches ambient 2×3. A coincidence surviving to $k = 3$ but failing at $k = 4$ would not be visible here. I report the domain rather than "small cases".

11 Instrument faults found this session

1. **A reflection implemented as a step-order reversal.** My group action reversed a tile's step sequence without negating the directions ($d \mapsto d + 3$). Reversing the order alone permutes a closed walk's edges into a *different* polygon — which still closes and is still counterclockwise, so neither my closure assertion nor my signed-area assertion detected it. The wrong code reported that the KTW set is reflection-closed for no g but that $\sigma_W(\mathsf{hex}) = \mathsf{hex}$; the correct code reports $\sigma_W(\mathsf{hex}) = \mathsf{hex}^{\mathsf{R}}$, which reverses the sign of the central argument in §6. The lesson in the form I want to keep it: *two invariants of a polygon that are both invariant under edge permutation cannot detect an edge permutation* — the control must be a property that distinguishes the orbit, here $\mathrm{act}(\mathrm{act}(x, g), g^{-1}) = x$ (now checked on all

204 (tile, g) pairs) plus an independent orientation test (does g send an up-triangle to an up-triangle?).

2. **A correct answer reached through a wrong intermediate.** Before fixing (1) I had written that the hexagon is wall-reflection-*invariant* and that this is "why clipping it in half is consistent". The clipping is consistent, and $\mathsf{trap} = \mathsf{hex} \cap H^+$ is true, but not for that reason. The true reason is Theorem 7: the wall passes through the centre, so the region splits $3 + 3$. Region-invariance and label-invariance are different statements and I had merged them.

3. **A relative-path emitter default.** `tracecheck.emit.init` defaults its log directory to a *relative* path, so my first call — made with cwd = `projects/` — wrote the session trajectory to `projects/state/trajectory/` instead of `state/trajectory/`. The file-listing check I ran immediately afterwards looked in the intended directory and showed only older sessions, which is what caught it. Re-pinned with an absolute `log_dir` and a stable session tag.

4. **A float where an integer was expected, harmlessly.** Engine B returns ± 1 sign factors as Python floats whenever the exponent $|\nu| - |\lambda| - |\mu|$ is negative (which is always, for d). The comparisons $4.0 == 4$ behave correctly, so no result is affected, but the printed values read "4.0" and a future reader should not take that as a non-integral structure constant.

12 What I would do next

The bijection in Gap 1 is the obvious next target and looks tractable: by Theorem 7 the only configurations differing between the two models are hexagons centred on the wall, and by Theorem 9 the clipping is label-preserving, so the content is entirely in showing that the enlarged puzzle's extra column is frozen. If that works it converts Proposition 18 from `computed` to `proved` and supplies the proof the paper omits.

The second target is the one I deliberately did not reach for: whether hex has an R-matrix provenance in the Wheeler–Zinn-Justin presentation rather than on its own faces. Theorem 4 is encouraging for this, because the vertex star is the natural face of the *dual* (honeycomb) lattice — and honeycomb is where Grassmannian puzzle integrability usually lives. That is a conjecture and is recorded as one.

References

[1] P. Pylyavskyy and J. Yang, *Puzzles in K-homology of Grassmannians*, arXiv:1801.07667. E-print read in full this session (`ktiles.tex`, `hex-tiling.tex`); all tile data, theorem statements, conventions and quotations above are from it.

[2] A. S. Buch, *A Littlewood–Richardson rule for the K-theory of Grassmannians.* Cited only for the definition of G_λ via set-valued tableaux, which I took from [1] lines 711–716 rather than from Buch directly; *not read this session.*

[3] M. Wheeler and P. Zinn-Justin, cited by [1] for the tri piece. *Not read this session*; no claim here depends on it.

[4] A. Knutson, T. Tao and C. Woodward, cited by [1] for the original puzzle pieces, which I extracted from [1] figure LR-tiles. *Not read this session.*